Yixin Wang

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Personal Profile

I am a Ph.D. student in Mathematics working on continuous optimization and its applications in machine learning and efficient AI. My research focuses on bilevel and structured optimization, Newton-type methods, and computationally efficient algorithms for large-scale problems.

Key Research Interests: Machine Learning, Efficient AI, Bilevel Optimization, Structured Optimization, Hyperparameter Selection, Optimal Damping, Newton-type Methods.

Education

Aug 2026 – Jul 2028: visiting Ph.D.
College of Computing and Data Science, Nanyang Technological University, Singapore
Honors: China Scholarship Council (CSC) Scholarship, 2025.
Supervised by Lim Wei Yang Bryan.
Research project: Low-Rank and Structured Optimization Methods for Resource-Aware Efficient AI (In Progress).

Sep 2023 – Jun 2029: Ph.D. in Mathematics (expected)
Beijing Institute of Technology, Beijing, China
Honors: The First Prize Scholarship (Beijing Institute of Technology, 2023 & 2024)
Supervised by Qingna Li.
Relevant Coursework: Modern Optimization Methods, Functional Analysis, Advanced Convex Optimization, Nonsmooth and Variational Optimization, Advanced Mathematical Statistics, Graph Theory.
Thesis: Optimization Methods for Bilevel Hyperparameter Selection in Machine Learning and for Optimal Damping (In Progress).

Sep 2019 – Jun 2023: B.Sc. in Mathematics **GPA:** 89.8/100
Jinan University, Guangdong, China
Honors: Outstanding Graduate (Jinan University, 2023), National Scholarship (Ministry of Education of China, 2022), The First Prize Scholarship (Jinan University, 2020, 2021 & 2022).
Relevant Coursework: Mathematical Analysis, Linear Algebra, Probability Theory, Mathematical Statistics, Optimization, Numerical Linear Algebra, Numerical Analysis, Mathematical modeling.
Thesis: Bilevel Hyperparameter Optimization Methods for Support Vector Classification — Based on Logistic Regression (LR) and the Biconjugate Gradient Method (BiCG).

Academic Publications

A fast smoothing Newton method for bilevel hyperparameter optimization for SVC with logistic loss.
Optimization, 2024, 74(12), 2793–2822. https://doi.org/10.1080/02331934.2024.2394612
***Yixin Wang**, Qingna Li.*

An efficient smoothing damped Newton method for large-scale bilevel hyperparameter optimization of SVC. Under review. arxiv preprint, 2026, arXiv:2506.22603v1
***Yixin Wang**, Qingna Li, Liwei Zhang*

A smoothing Newton method for optimal damping. In preparation, 2026.
***Yixin Wang**, Qingna Li, Françoise Tisseur*

Academic Activities

SIAM Conference on Optimization (OP26)
Invited Presentation: Smoothing Damped Newton Method for Large-Scale Mathematical Programs with Equilibrium Constraints (The University of Edinburgh, June 2 – 5, 2026)

EUROPT 2025 Summer School – The 5th edition of the EUROPT Summer School
Participant (University of Southampton, June 27 - 28, 2025)

EUROPT 2025 – The 22nd EUROPT Conference on Advances in Continuous Optimization
Invited Presentation: A highly efficient single-loop smoothing damped Newton method for large-scale bilevel hyperparameter optimization of SVC (University of Southampton, June 29 - July 2, 2025)

Summer School on Mathematical Communication Skills Workshop
Participant (The University of Manchester, Sep 8 - 12, 2025)

Skills

MATLAB, Python, C/C++.
Languages: English (Fluent) IELTS: 7.0, CET-4: 578, CET-6: 536